

Ezra Miller Combined Plot

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R Markdown

Plot for The final query in the .sql file. Combines 3 different time series: monthly revenue, monthly movie frequency (how many movies Ezra Miller was involved in), and monthly search interest. Each of these values were scaled by dividing its current value with the max value, transformed into a percentage. For example, if Ezra Miller was in 1 movie during the month of June 2016, then that scaled value would be 0.5 since the max amount of movies he was involved at a time was 2 ($1/2 * 100$).

```
library(DBI)
library(RMariaDB)
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##   filter, lag
```

```
## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
```

```
# Create connection object
```

```
con <- dbConnect(
  RMariaDB::MariaDB(),
  host      = "localhost",      # Or "localhost" if the server is on your machine
  port      = 3306,             # Default MySQL port
  user      = "root",           # Usually "root" for local setups
  password  = "sesame80",       # do not put password in here. Github etc.
  dbname    = "sem_project"
)
```

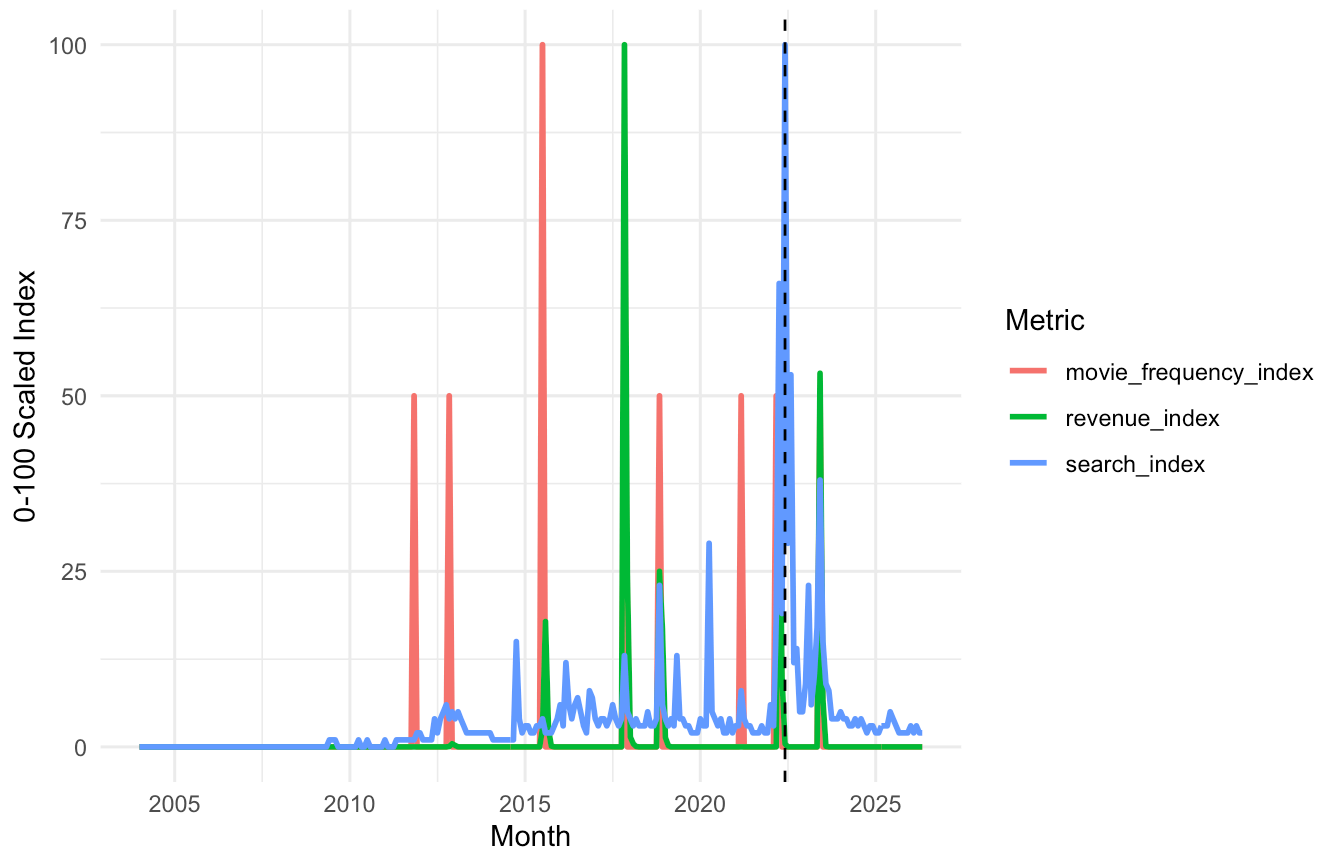
```
library(ggplot2)
```

```
plot_data <- dbGetQuery(con, "WITH monthly_search AS (
  SELECT
    DATE_FORMAT(trend_date, '%Y-%m-01') AS month_start,
    AVG(search_interest) AS avg_search_interest
  FROM ezra_search_trends
  GROUP BY month_start
), monthly_revenue AS (
  SELECT
    STR_TO_DATE(DATE_FORMAT(revenue_date, '%Y-%m-01'), '%Y-%m-%d') AS month_start,
    SUM(daily_revenue) AS total_monthly_revenue
  FROM ezra_daily_revenue
  GROUP BY month_start
), monthly_releases AS (
  SELECT
    DATE_FORMAT(release_date, '%Y-%m-01') AS month_start,
    COUNT(*) AS movies_released
  FROM ezra
  GROUP BY month_start
), combined AS (
  SELECT
    s.month_start,
    s.avg_search_interest,
    COALESCE(r.total_monthly_revenue, 0) AS total_monthly_revenue,
    COALESCE(m.movies_released, 0) AS movies_released
  FROM monthly_search s
  LEFT JOIN monthly_revenue r
    ON s.month_start = r.month_start
  LEFT JOIN monthly_releases m
    ON s.month_start = m.month_start
)
SELECT
  month_start,
  avg_search_interest,
  total_monthly_revenue,
  movies_released,
  ROUND(avg_search_interest / MAX(avg_search_interest) OVER () * 100, 2) AS search_index,
  ROUND(total_monthly_revenue / MAX(total_monthly_revenue) OVER () * 100, 2) AS revenue
```

```
e_index,  
  ROUND(movies_released / MAX(movies_released) OVER () * 100, 2) AS movie_frequency_in  
dex  
FROM combined  
ORDER BY month_start;")  
  
plot_data$month_start <- as.Date(plot_data$month_start)  
  
plot_long <- plot_data %>%  
  pivot_longer(  
    cols = c(search_index, revenue_index, movie_frequency_index),  
    names_to = "metric",  
    values_to = "index_value"  
  )  
  
ggplot(plot_long, aes(x = month_start, y = index_value, color = metric)) +  
  geom_line(linewidth = 1) +  
  geom_vline(xintercept = as.Date("2022-06-01"), linetype = "dashed") +  
  labs(  
    title = "Ezra Miller: Search Attention vs Box Office Activity",  
    subtitle = "Metrics are normalized to a 0-100 index for comparison",  
    x = "Month",  
    y = "0-100 Scaled Index",  
    color = "Metric"  
  ) +  
  theme_minimal()
```

Ezra Miller: Search Attention vs Box Office Activity

Metrics are normalized to a 0–100 index for comparison



Note that the `echo = FALSE` parameter was added to the code chunk to prevent printing of the R code that generated the plot.